# Proxy Features and Fairness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

## One readable proxy-and-ablation evidence record.

Trace retained information,compare matched errors and request missing contextual evidence.

**Keep one habit:** W1–W3/W5 → selected new evidence request → project proxy record → changed-case return.

MAIN LESSON · SECTION 1

## Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Subgroup Performance. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

### Subgroup Performance

> **My remembered idea, example and correction**

### Feature Engineering

> **My remembered idea, example and correction**

### Reproducible Experiments

> **My remembered idea, example and correction**

## W1 · Explain the mechanism

> Keep this task’s complete inputs, stated rule and support record together.

### Task and evidence record

- MainNorthbuildingX8/10, SouthX2/10; the 20-case association has counterexamples and proves no causal harm. PicturecameraG/Dis a perfect setting identifier only in that original toy sample; new crossed assignments need not preserve it. Complete ablation trainingT1/T2 allbits1, T3/T4 allbits0; settingmetadataexcluded. V1 chooses camera_codeby training tie, V2 removes it and chooses room_hint.

  ValidationP1 marker1/P2 marker0 each4 camera/settingviews:both 4/8 correct. V1 settingrecall1/2 each, V2 greenhouse2/2 desk0/2; two new plants, not 8 independent plants. No real fairness or disease conclusion.

- What do the device counts support, and what do they not support?

> **My reasoning and evidence**

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MAIN LESSON · SECTION 13

## W2 · Trace the current experiment

> Keep this task’s complete inputs, stated rule and support record together.

### Task and evidence record

- MainNorthbuildingX8/10, SouthX2/10; the 20-case association has counterexamples and proves no causal harm. PicturecameraG/Dis a perfect setting identifier only in that original toy sample; new crossed assignments need not preserve it. Complete ablation trainingT1/T2 allbits1, T3/T4 allbits0; settingmetadataexcluded. V1 chooses camera_codeby training tie, V2 removes it and chooses room_hint.

  ValidationP1 marker1/P2 marker0 each4 camera/settingviews:both 4/8 correct. V1 settingrecall1/2 each, V2 greenhouse2/2 desk0/2; two new plants, not 8 independent plants. No real fairness or disease conclusion.

- Why might removing building but keeping device code leave related information?

> **My reasoning and evidence**

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MAIN LESSON · SECTION 14

## W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- MainNorthbuildingX8/10, SouthX2/10; the 20-case association has counterexamples and proves no causal harm. PicturecameraG/Dis a perfect setting identifier only in that original toy sample; new crossed assignments need not preserve it. Complete ablation trainingT1/T2 allbits1, T3/T4 allbits0; settingmetadataexcluded. V1 chooses camera_codeby training tie, V2 removes it and chooses room_hint.

  ValidationP1 marker1/P2 marker0 each4 camera/settingviews:both 4/8 correct. V1 settingrecall1/2 each, V2 greenhouse2/2 desk0/2; two new plants, not 8 independent plants. No real fairness or disease conclusion.

- Describe a controlled with/without-code comparison.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 16

## W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- MainNorthbuildingX8/10, SouthX2/10; the 20-case association has counterexamples and proves no causal harm. PicturecameraG/Dis a perfect setting identifier only in that original toy sample; new crossed assignments need not preserve it. Complete ablation trainingT1/T2 allbits1, T3/T4 allbits0; settingmetadataexcluded. V1 chooses camera_codeby training tie, V2 removes it and chooses room_hint.

  ValidationP1 marker1/P2 marker0 each4 camera/settingviews:both 4/8 correct. V1 settingrecall1/2 each, V2 greenhouse2/2 desk0/2; two new plants, not 8 independent plants. No real fairness or disease conclusion.

- Does unchanged accuracy after removing a proxy prove **fairness**?

> **My reasoning and evidence**

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MAIN LESSON · SECTION 4

## Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- The selected postcode/school-code task gives a claimed removal, not actual association counts, model predictions or a mapping. Do not infer any real student's location or identity. State the unsupported part, request provenance/association/context evidence, and propose matched ablations and group error checks. No automatic fairness or complete-removal result follows without those observations.

- A team removes postcode but retains school code and claims all location-related information is gone. Evaluate the claim and request **evidence**.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Independent case: Apply independently

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

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MAIN LESSON · SECTION 9

## Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Document Plant Classifier proxy risks, paired-test **evidence**, **mitigation**, and remaining **fairness** limitation.

- The project change and **evidence** I will keep

**Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

# Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

MAIN LESSON · SECTION 10

# Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

## Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Use W2: Find a proxy risk for **Proxy feature**s and **fairness**. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

## Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

### Task and evidence record

- Use the worked case for **Proxy feature**s and **fairness**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

## Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

### Task and evidence record

- A leaf model performs well using location ID alone. Explain a possible proxy problem and design a diagnostic evaluation.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

## Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Background colour tracks collection location, associated with plant types.
- Explain proxy risk and test.
- Show your trace, calculation or **evidence**

**Write or draw your reasoning and evidence.**

▼ Extra practice · Changed case

REFERENCE · SECTION 7

## Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- A plant classifier uses camera_model, and one species was photographed mostly on a single camera. What shortcut might it learn and how could you investigate?

- My independent answer and reason

> **Write or draw your reasoning and evidence.**

**Week 32 Test and reflect**

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

## Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

## Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

### Task and evidence record

- A leaf model performs well using location ID alone. Explain a possible proxy problem and design a diagnostic evaluation.

- Record support used: none, hint or teacher explanation.

**My independently reasoned answer**

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

# Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- A team removes postcode but retains school code and claims all location-related information is gone. Evaluate the claim and request **evidence**.

**My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

# Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- Removing an obvious setting field may leave another field that carries the same information.
- Toy table has location=greenhouse/desk and camera_code=G/D. Every G record is greenhouse; every D record is desk. The team drops location.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- What **evidence** contradicts the original "all setting information removed" claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/32

# Proxy Features and Fairness

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.

### 1. Remove direct field

The location column is absent from the new input table.

### 2. Check retained proxy

camera_code still identifies the setting perfectly in this supplied sample.

### 3. Audit effect

Compare controlled predictions and held-out setting slices; dropping one column does not establish setting independence.

**CHANGE ONE THING**

Camera codes are assigned independently of setting in a new balanced dataset.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

**My picture-based explanation and changed-case prediction**

REFERENCE · SECTION 19

# Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

## Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

# Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

## First supported question and hint

- Removing an obvious setting field may leave another field that carries the same information.
- Which labelled step matches this detail? Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.
- 1. Remove direct field
- 2. Audit effect
- 3. Check retained proxy
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Audit effect Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence. Read the picture in order.

Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

## Changed-case question and hint

- Changed-case practice: Camera codes are assigned independently of setting in a new balanced dataset. Which conclusion is supported by the supplied facts?
- 1. camera_code no longer perfectly identifies setting in that dataset.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: camera_code no longer perfectly identifies setting in that dataset. The observed mapping changed; other retained fields may still contain setting information.

## Repair and keep the record

Review Subgroup Performance (ai-4/31). Goal: Calculate subgroup results with sample sizes. Short practice: Why should the report retain subgroup rates alongside the total?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Feature Engineering (ai-4/29). Goal: Create features available at **prediction** time. Short practice: What should be saved so another group transforms the query identically?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean **70%**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Keep your completed work**

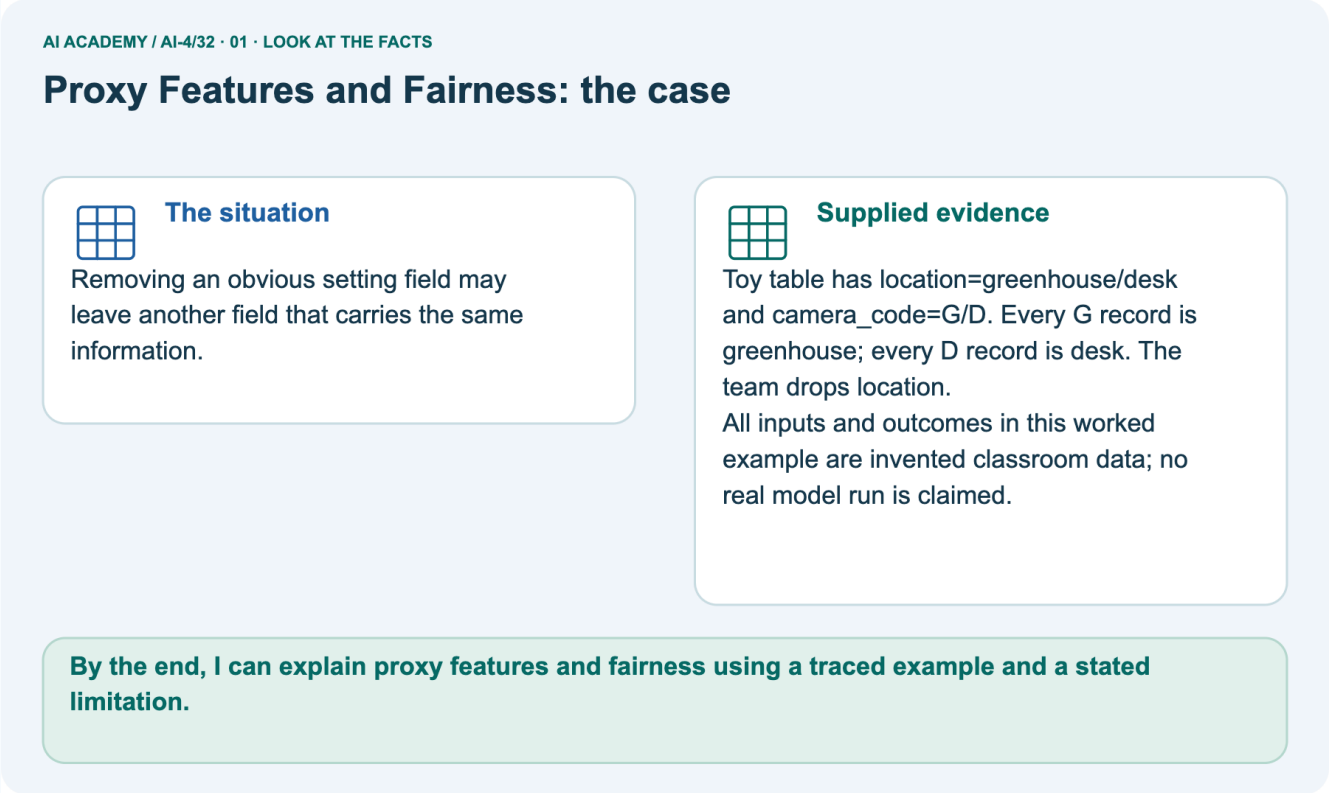
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

## Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

## 01. Proxy Features and Fairness: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

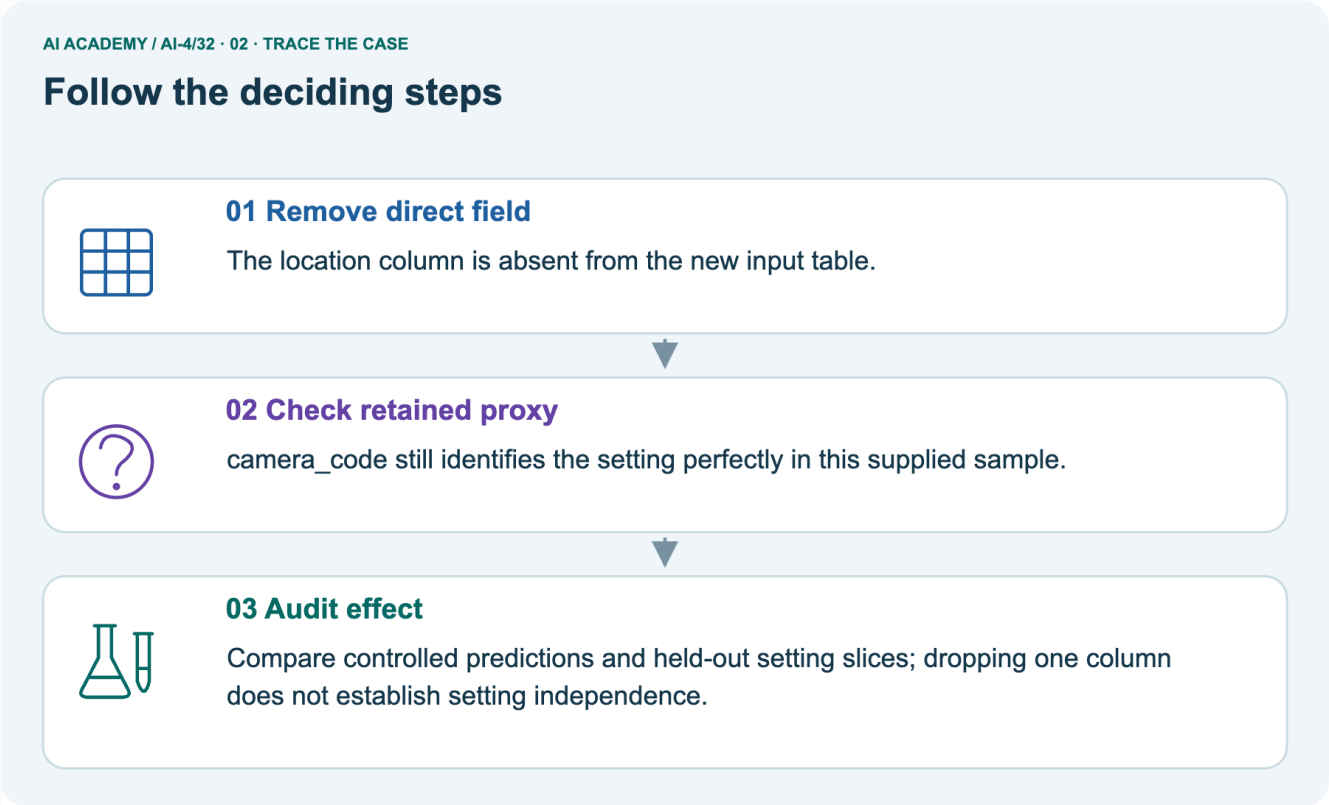
Removing an obvious setting field may leave another field that carries the same information. Supplied facts: Toy table has location=greenhouse/desk and camera_code=G/D. Every G record is greenhouse; every D record is desk.

The team drops location. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps



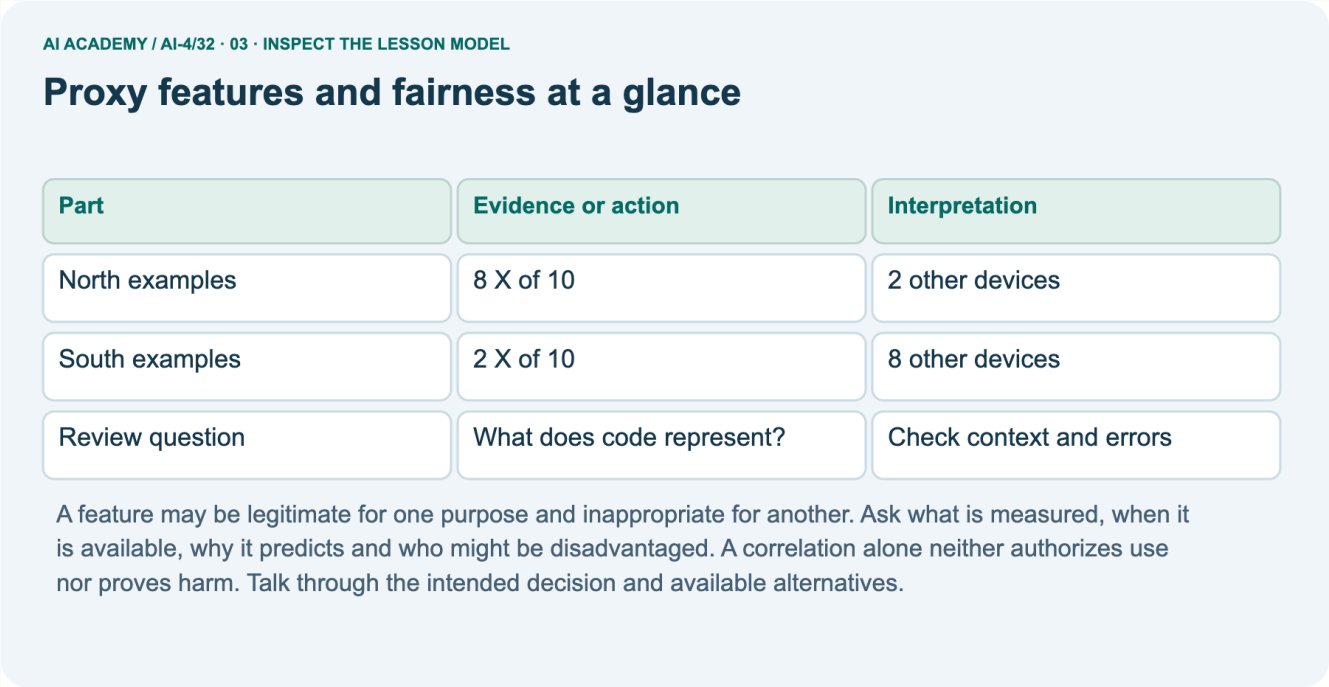

Follow the labelled picture with the matching explanation. Open the vector diagram

Remove direct field: The location column is absent from the new **input** table. → Check retained proxy: camera_code still identifies the setting perfectly in this supplied sample. → Audit effect: Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. Proxy features and fairness at a glance



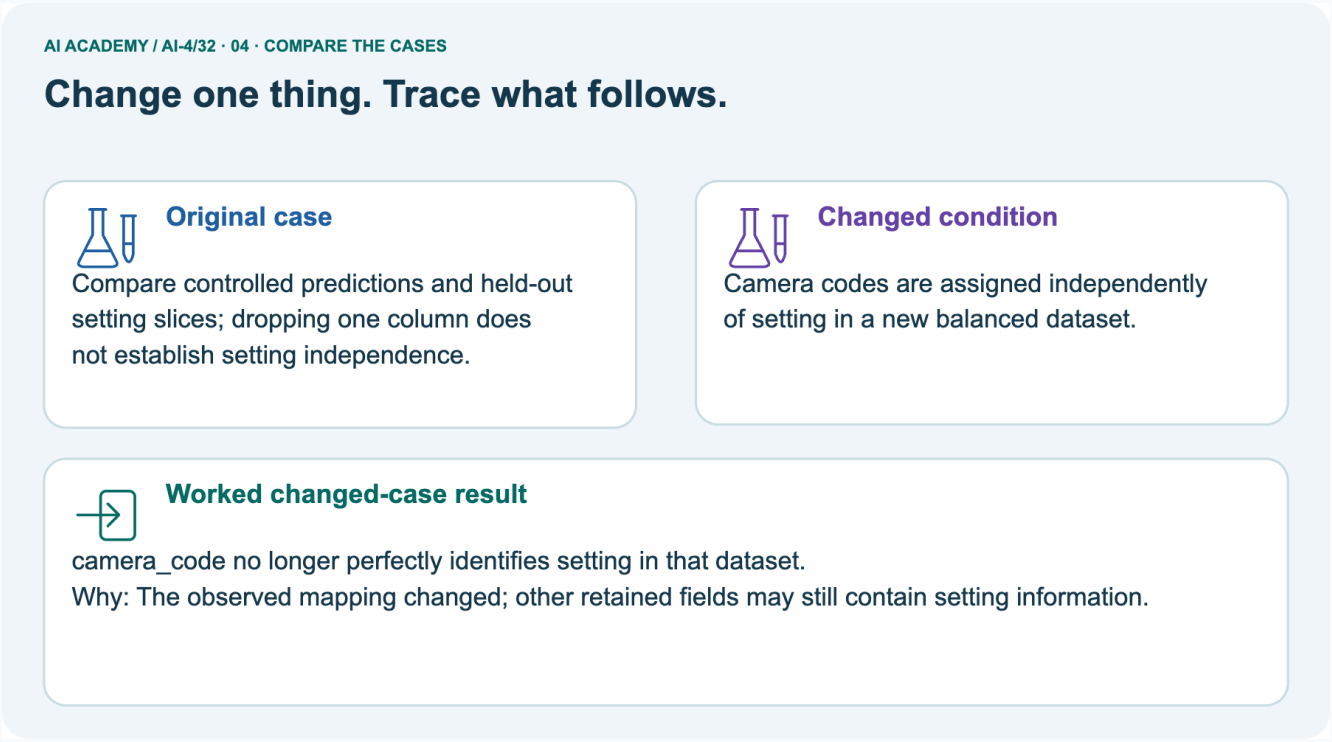

Follow the labelled picture with the matching explanation. Open the vector diagram

**Proxy feature**s and **fairness** at a glance. North examples | 8 X of 10 | 2 other devices. South examples | 2 X of 10 | 8 other devices. Review question | What does code represent? | Check context and errors

Read the labels and values. Explain the relationship the model shows.

### My picture explanation, deciding evidence and one limit

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

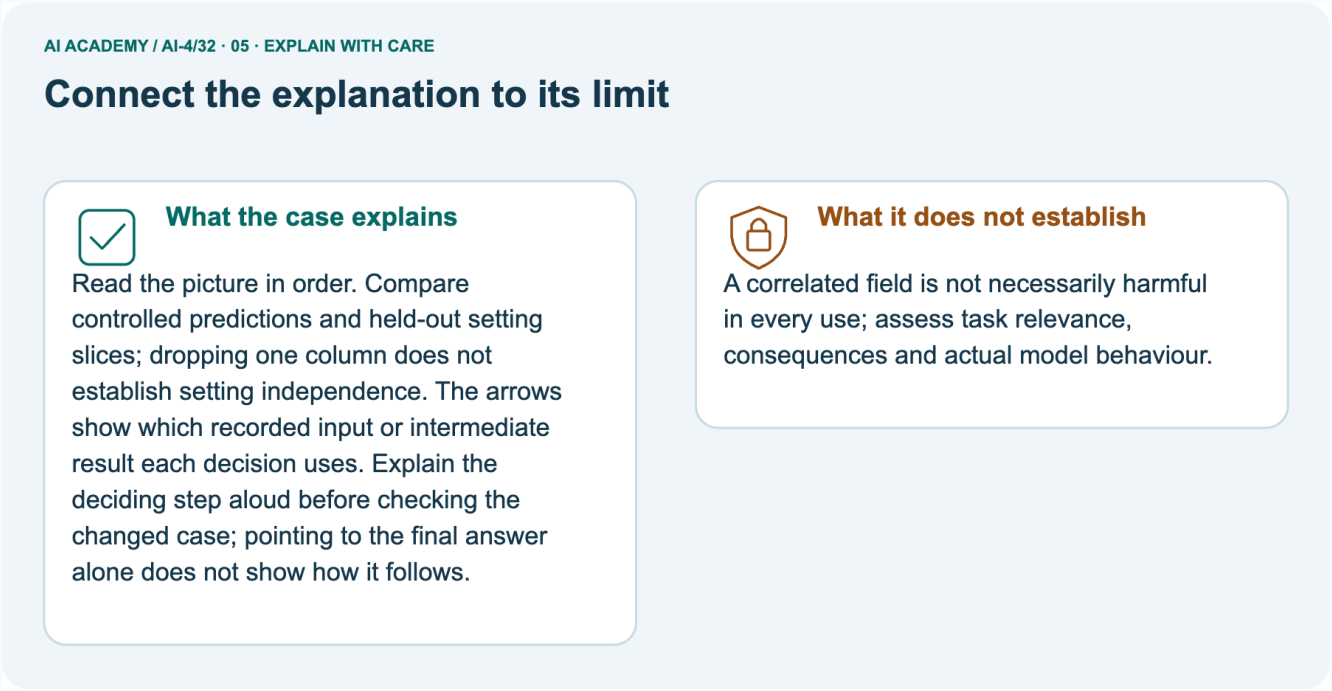
Changed condition: Camera codes are assigned independently of setting in a new balanced dataset. Expected result: camera_code no longer perfectly identifies setting in that dataset.

Reason: The observed mapping changed; other retained fields may still contain setting information.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

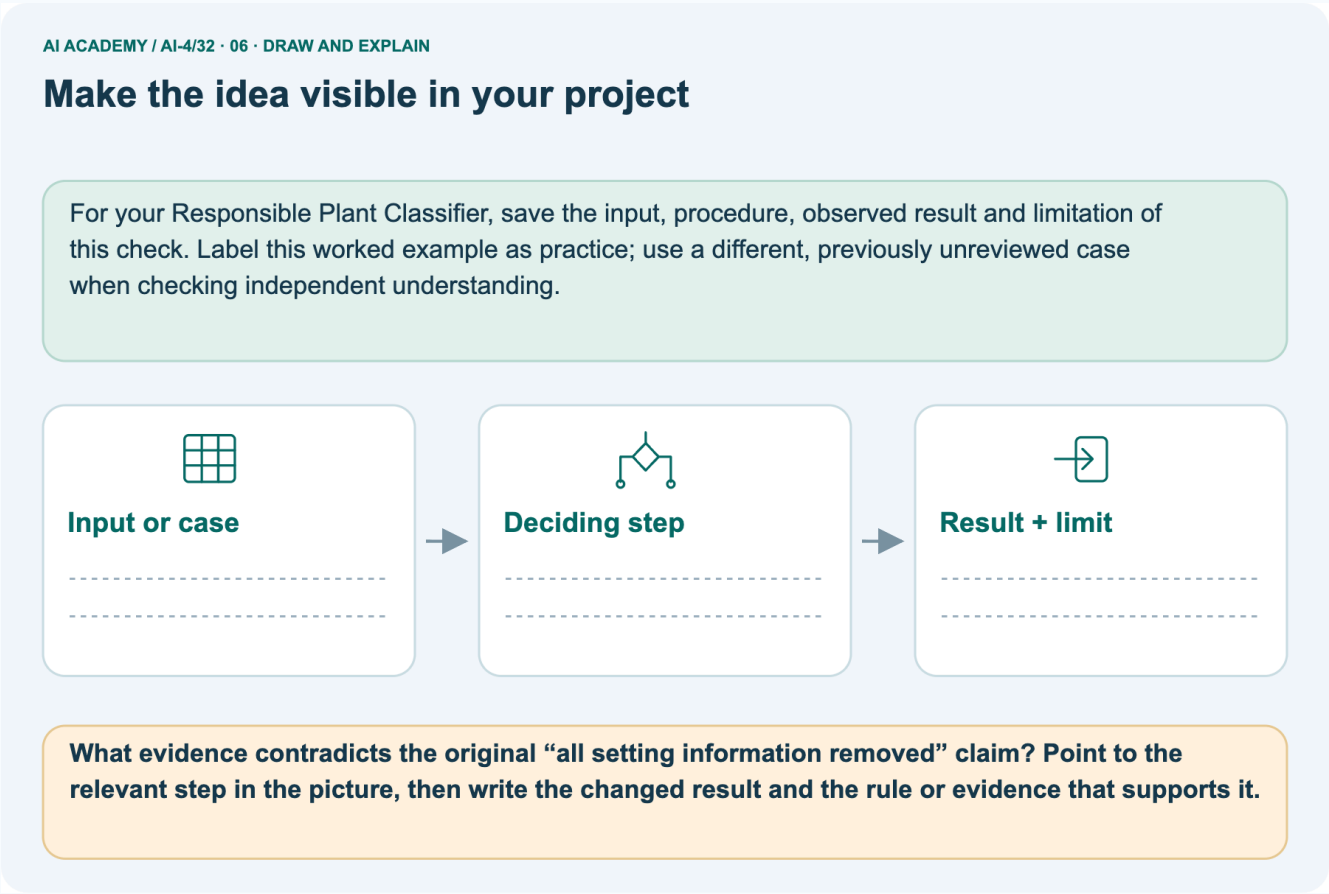
The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A correlated field is not necessarily harmful in every use; assess task relevance, consequences and actual model behaviour.

Explain the case, then point to a limit on the claim.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What **evidence** contradicts the original “all setting information removed” claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

**My picture explanation, deciding evidence and one limit**

▼ Earlier wording, alternatives and picture transcripts

**01. Proxy Features and Fairness: the case**

**02. Follow the deciding steps**

**03. Proxy features and fairness at a glance**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.